

Chapter 7

Expansion Bus and Cards

The components inside the computer talk to each other using a path that carries data. Most internal components including the processor, memory, expansion cards, storage devices, talk to each other over a bus. Bus is simply a channel over which information flows between two or more devices. There are different types of bus:

- memory bus
- cache bus
- processor bus
- I/O bus(expansion bus)

Memory Bus

- used to connect chipset to RAM
- is second level bus

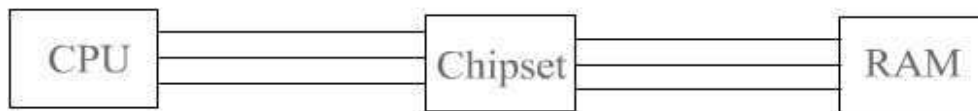


fig memory bus

Cache Bus

This is the bus that connects the processor (CPU) with the cache memory. It is also called Back Side Bus unlike Front Side Bus. It is high level bus.

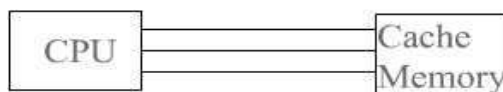


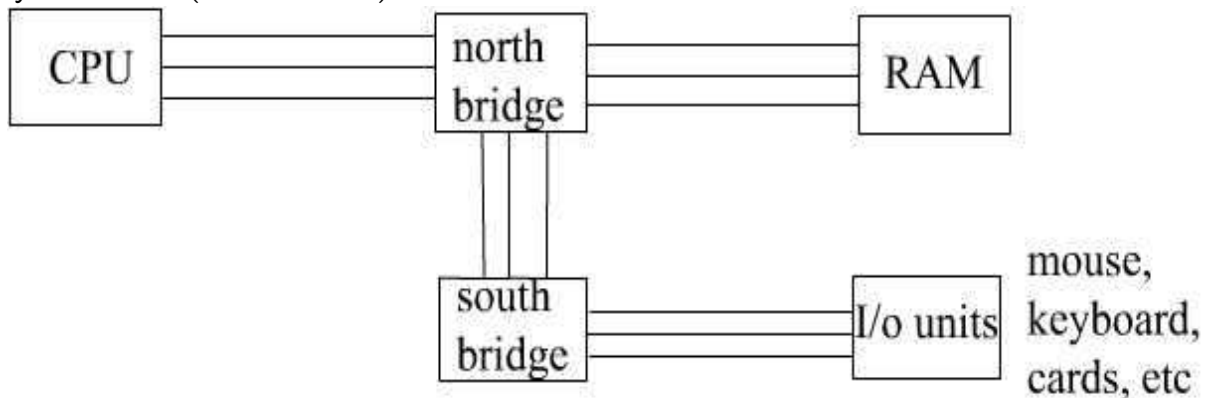
fig cache bus

Processor Bus

It is the highest level bus. It is used to send information to and from the processor.

Expansion Bus

Expansion bus provide connection between internally connected peripherals and system board(motherboard).



There are different types of I/O buses. These are:

- ISA(Industry standard architecture)
- EISA(Extended ISA)

- MCA(Micro Channel Architecture)
- PCI(Peripheral Component Interconnect)
- VESA(Video electronics Standard Association)
- USB(Universal Serial Bus)

Most modern PCs all have basic peripheral devices built into the motherboard. Still, we want to add additional devices to our PC like sound card, network card, modem, video adapter, SCSI card, etc. These cards are installed on expansion buses on the motherboard.

i) ISA

ISA stands for Industry Standard Architecture. It was introduced in 1981 with the IBM PC. There are two types of ISA bus:

- 8 bit ISA bus
- 16 bit ISA bus

8 bit ISA Bus:

- provides eight data lines
- runs at 4.77 MHZ
- introduced with the first IBM PC

16 bit ISA Bus:

- provides 16 bit data bus
- runs at 8MHZ(8.33 MHZ)
- introduced by IBM in 1984

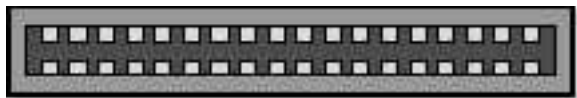


fig ISA bus layout

ISA bus is black in color.

In general, ISA buses have problems:

- slow in terms of speed
- configuration of cards is difficult-setting IRQ and other resources
- No intelligence like plug and play-manual set up, not automatic

ii) MCA

- was introduced by IBM in 1986. The aim was to replace the ISA bus with something bigger and better.
- 32 bits wide
- it is proprietary
- It offers several significant improvement over ISA
- It is intelligent and self configuring(plug and play)
- supports speeds of up to 10.33MHZ
- incompatible with ISA. This affected its acceptance in the market

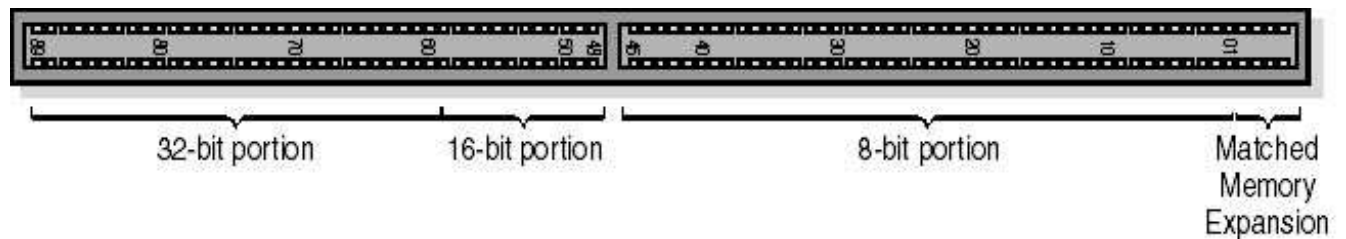


fig MCA channel lay out

iii) EISA

- stands for extended ISA. It was introduced by Compaq as a response to the introduction of MCA by IBM- because of the proprietary nature of MCA. It was introduced in 1988-89.
- provides 32 bit bus width and 8.33MHZ speed
- It is intelligent and self configuring
- It is compatible with ISA and it is non proprietary. You can install ISA card on EISA bus(compatible).

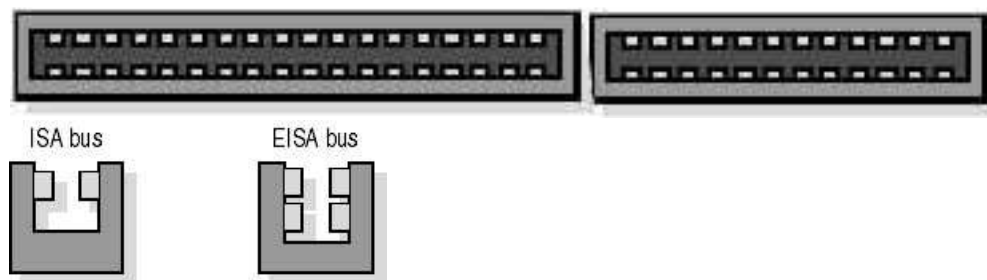


fig EISA bus

iv) Local Buses

The I/O buses discussed so far are relatively slow in speed. As the need better speed GUI operating system and graphics applications arises, the need for better speed bus was unquestionable.

To solve the problem of speed, engineers developed the local bus. Local bus connects special peripherals to the system board through expansion slot and enable the peripherals to work at the speed close to the microprocessor.

The major local buses are:

- VESA (Video Electronics Standard Association)
- PCI(Peripheral Component Interconnect)
- AGP(Advanced Graphics Port)

1) VESA

- introduced in 1992 by a group called VESA
- it was introduced to improve video performance. It was developed for video adapters(cards).
- It moves data 32 bits at a time, 33 MHZ speed
- It worked best in 486 processors. It is obsolete today.

2) PCI

-introduced in 1992 by Intel. It has the same goals as VESA, solving speed bottlenecks. It is the most popular today.

It has a lot of benefits:

- high performance (the highest performance I/O bus)
- low cost
- automatic configuration (plug and play)
- it runs at the speed of 33MHZ. Its advantage over VESA lies in the chipset that controls it. It has a special chipset in the circuitry that is designed to control it.
- it supports a wide variety of cards compared to VLB(VESA). It supports video cards, SCSI adapters, high speed network cards, etc. Hard disk drives are also connected to PCI bus on the motherboard.
- PCI slots are white in color.

3) AGP

- created by Intel as new bus specifically for high performance graphics and video support.
- It is based on PCI but contains a number of improvement over PCI. It is brown in color.
- was developed in response to greater and greater performance requirements for video.
- runs at speed of 66MHZ (double that of PCI) base clock rate
- AGP will allow the speed of video card to pace the requirements of high speed 3D graphics rendering and full motion video on the screen.

4) USB

- stands for Universal Serial Bus
- used to connect external peripheral devices like keyboard, mouse, printer, scanner, etc.
- You can connect any device to USB port and it supports it.

Configuring Expansion Cards

External and internal computer hardware such as disk drives, monitors, and internet lines can be connected to computer motherboard by using expansion cards. The most common cards connected to expansion slots include:

- Network Interface Card(NIC)-for network
- Internal modem-for dial up internet
- Graphics card- for video display(monitor control)
- TV card- to display TV on computer
- Sound Card-for sound processing
- SCSI Card-connect devices to computer

All cards installed in the expansion slots need three things to function properly:

- I/O address (Input Output Address)
- IRQ (Interrupt Request)
- DMA (Direct Memory Access)

I/O Address

Bus system establish a connection between CPU and expansion devices(cards). They provide a path for flow of data. But the CPU has to know which device is sending data and which device should receive data. This is done using I/O address.

I/O address is a unique ID given to devices for identification purpose. Everything in computer, hardware or software, requires a unique name and address for the CPU to be able to identify what is going on.

The CPU must identify the device before any data is placed on the bus. CPU uses two wires(bus wires) to notify the devices that the address bus is not being used to specify memory address, but rather to read or write to a particular device:

- Input/Output Read(IOR)

- Input/Output Write(IOW)

Though address bus has more than 20 wires(20 bits), when IOW or IOR has voltage, only 16 wires (16 bits) are relevant (monitored). These 16 wires are used to send the ID of the device that is involved in the communication. These wires carry I/O addresses of a device in another word.

- no two devices can have the same I/O address

- I/O addresses are written in the form of hexadecimal number

- I/O address is written in capital letters(case sensitive)

- 0 is the first number in I/O addresses and it is ignored(you don't have to write it)

On AT motherboard, the following input output addresses are available:

- 220h-26Fh

- 320h-32Fh

You can use these addresses for configuring expansion cards.

Configuration:

- using jumpers

- using switches

- using BIOS

IRQ(Interrupt Request)

The CPU does a lot of activities like running application programs, running the operating system, etc. It has no time to give full attention to devices all the time. So when a device needs CPU attention, it sends a request to the CPU. Then the CPU stops what it does and pays attention to

Every CPU has a wire called interrupt(INT). If voltage is applied to this wire, the CPU interrupts what it is doing and attends to the device. E.g. mouse click causes interruption of what CPU does and listen to it. But CPU has only one INT wire, but it handles many peripheral devices. A CPU is helped by a chipset to identify which device is interrupting. This chipset is called 8259 chip. Every device that needs to interrupt CPU is provided with a wire called IRQ.

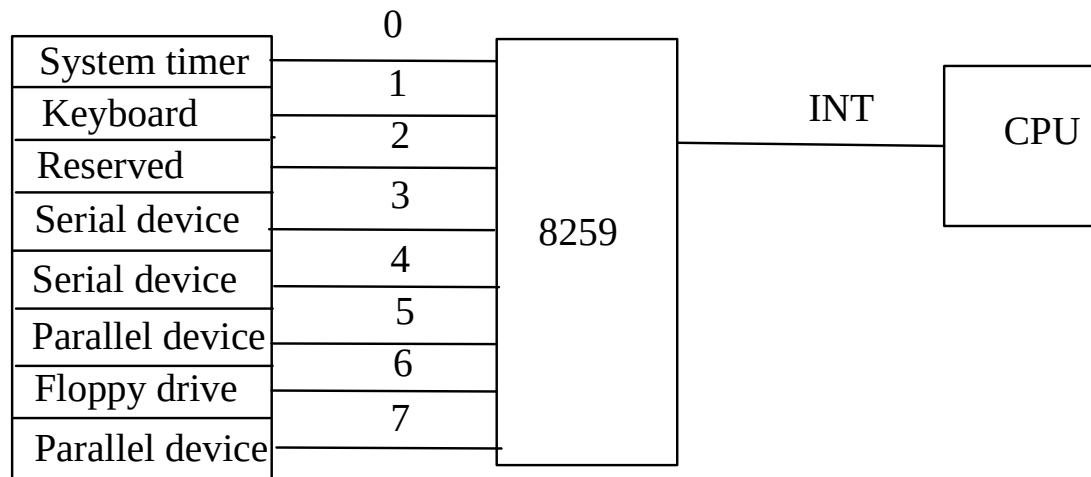


Fig interrupt number

The process of interrupt is:

- 1) Device applies voltage to 8259 chip through IRQ wire
- 2) 8259 informs the CPU that interrupt is pending through INT wire.
- 3) CPU uses wire called INTA to signal 8259 chip to send the I/O address of the device. This enables the CPU know what device wants to talk to him.
- 4) the CPU runs BIOS related to the device

8088-one 8259 chip

80286-two 8259 chip

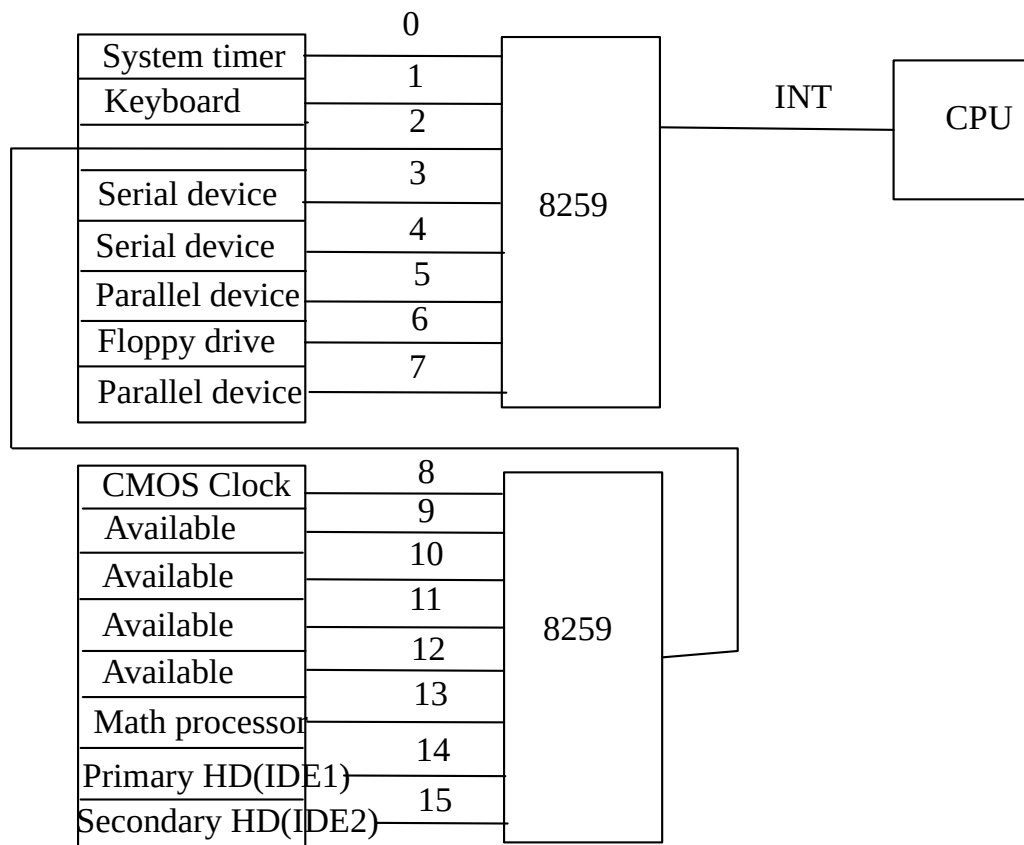


Fig Cascading 8259 chips

The cascading removes IRQ 2. IRQ 9 is directed to the old IRQ 2 wire. Because of this IRQ 2 and IRQ 9 are the same.

IRQs 2/9, 10, 11, and 12 are available for you to use for configuration. The rest are preoccupied though some of them can still be used. IRQs 3, 4, 5, 7, and 15 could be changed and used for other devices listed above.

IRQ 3- COM2, COM4

IRQ 4-COM1, COM3

IRQ5-LPT2

IRQ 7-LPT1

IRQ 15-secondary IDE

-no two devices can have the same IRQ like I/O address. To avoid conflict, document the IRQs you set for each device you install. This works for I/O address also.

Configuration:

- using jumpers
- using switches
- using BIOS

Direct Memory Access(DMA)

CPU has to move a lot of data using considerable CPU time and power for what is a simple task. This is wastage of CPU resource. To reduce this wastage, a chip called DMA was installed to work with CPU. The purpose of this chip is to move data. It handles data movement from peripherals to RAM and from RAM to peripherals. It is numbered as 8237 chip.

Not all devices use DMA. The following devices use DMA:

- few SCSI controllers
- Sound Cards
- Network Cards
- Some CD-ROM drives

No two devices can use the same DMA channel. DMA conflicts cause problem in the PC.

<u>DMA Channel</u>	<u>Function</u>
0	available
1	available
2	Floppy disk controller
3	ECP parallel/available
4	First DMA controller
5	second sound card
6	SCSI/available
7	available

Configuration:

- use jumpers
- use switches

-use BIOS

To install any non-plug and play card:

- first read device manual
- assign unique DMA, I/O address, and IRQ
- document these configurations for future use

To install plug and play card:

- install the device(insert it into the motherboard)
- turn on the PC and use it

COM and LPT Ports

Serial and parallel devices have preset combinations of IRQ and I/O addresses. The preset combinations are called ports. The purpose of port is to make installation easier. Devices like external modems, printers do not require IRQ or I/O addresses.

<u>Port</u>	<u>I/O address</u>	<u>IRQ</u>
COM1	3F8	4
COM2	2F8	3
COM3	3E8	4
COM 4	2E8	3
LPT1	378	7
LPT2	278	5